14. Factor Groups

1. $\langle 3\rangle$ has 2 elements, so $\mathbb{Z}_6/\langle 3\rangle$ has $6/2 = 3$ elements.

2. $\langle 2\rangle \times \langle 2\rangle$ has $2 \cdot 6 = 12$ elements, so the factor group has $48/12 = 4$ elements.

3. $\langle (2,1)\rangle$ has 2 elements, so the factor group has $8/2 = 4$ elements.

4. $\{0\} \times \mathbb{Z}_5$ has 5 elements, so the factor group has $15/5 = 3$ elements.

5. $\langle (1,1)\rangle$ has 4 elements, so the factor group has $8/4 = 2$ elements.

6. $\langle (4,3)\rangle$ has 6 elements, so the factor group has $(12 \cdot 18)/6 = 36$ elements.

7. $\langle (1,\rho_1)\rangle$ has 6 elements, so the factor group has $12/6 = 2$ elements.

8. (1, 1) generates the entire group so the factor group has just one element.

9. $\langle 4\rangle = \{0, 4, 8\}$. Now $5 + 5 = 10$, $5 + 5 + 5 = 3$, and $5 + 5 + 5 + 5 = 8$. Because $5 + 5 + 5 + 5 = 8$ is the first repeated sum of 5 in $\langle 4\rangle$, we see that the coset $5 + \langle 4\rangle$ is of order 4 in this factor group.

10. $\langle 12\rangle = \{0, 12, 24, 36, 48\}$. We prefer to compute sums of the element 2 in the coset $26 + \langle 12\rangle$, rather than the element 26. Computing, $2 + 2 = 4$, $4 + 2 = 6$, $6 + 2 = 8$, $8 + 2 = 10$, and $10+2 = 12 \in \langle 12\rangle$. Thus $26 + \langle 12\rangle$ has order 6 in the factor group.

11. $\langle (1,1)\rangle = \{(0,0), (1,1), (2,2), (0,3), (1,4), (2,5)\}$. Computing, (2, 1) + (2, 1) = (1, 2), (1, 2) + (2, 1) $= (0,3) \in \langle (1,1)\rangle$. Thus $(2,1) + \langle (1,1)\rangle$ has order 3 in the factor group.

12. $\langle (1,1)\rangle = \{(0,0), (1,1), (2,2), (3,3)\}$. Computing, we find that (3, 1) + (3, 1) $= (2,2) \in \langle (1,1)\rangle$. Thus $(3,1) + \langle (1,1)\rangle$ has order 2 in the factor group.

13. $\langle (0,2)\rangle = \{(0,0), (0,4), (0,6)\}$. Computing, (3, 1) + (3, 1) = (2, 2), (2, 2) + (3, 1) = (1, 3), (1, 3) + (3, 1) $= (0,4) \in \langle (0,2)\rangle$. Thus $(3,1) + \langle (0,2)\rangle$ has order 4 in the factor group.

14. $\langle (1,2)\rangle = \{(0,0), (1,2), (2,4), (3,6)\}$. We have $(3,3) + (3,3) = (2,6), (2,6) + (3,3) = (1,1), (1,1) + (3,3) = (0,4), (0,4) + (3,3) = (3,7), (3,7) + (3,3) = (2,2), (2,2) + (3,3) = (1,5), (1,5) + (3,3) = (0,0) \in \langle (1,2)\rangle$. Thus $(3,3) + \langle (1,2)\rangle$ has order 8 in the factor group. It generates the entire factor group.

15. $\langle (4,4)\rangle = \{(0,0), (4,4), (2,0), (0,4), (4,0), (2,4)\}$. We see that $(2,0) \in \langle (4,4)\rangle$ so $(2,0) + \langle (4,4)\rangle$ has order 1 in the factor group.

16. Because $\rho_1\mu_1\rho_1^{-1} = \rho_1\mu_1\rho_2 = \mu_2$ and $\rho_1\rho_0\rho_1^{-1} = \rho_1\rho_0\rho_2 = \rho_0$, we see that $i_{\rho_1}(H) = \{\rho_0, \mu_2\}$.

17. The definition is incorrect.

A **normal subgroup** H of a group G is a subgroup satisfying $gH = Hg$ for all $g \in G$.

18. The definition is correct.

19. The definition is incorrect. Change "homomorphism" to "isomorphism" and "into" to "onto".

An **automorphism** of a group G is an isomorphism mapping G onto G.

20. Normal subgroups are those whose cosets can be used to form a factor group, because multiplication of left cosets by multiplying representatives is a well-defined binary operation.

21. (See the answer in the text for Part(**a**) and Part(**b**)).

c. Taking a and b as representatives of the cosets aH and bH respectively, we see that $(aH)(bH) = (ab)H$. Because G is abelian, $ab = ba$, so $(ab)H = (ba)H = (bH)(aH)$. Thus $(aH)(bH) = (bH)(aH)$ so G/H is abelian.

22. **a.** When working with a factor group G/H, one would let x be an element of G, not an element of G/H. The student probably does not understand what elements of G/H look like and can write nothing sensible concerning them.

b. We must show that each element of G/H is of finite order. Let $xH \in G/H$.

c. Because G is a torsion group, we know that $x^m = e$ in G for some positive integer m. Computing $(xH)^m$ in G/H using the representative x, we have $(xH)^m = x^m H = eH = H$, so xH is of finite order. Because xH can be any element of G/H, we see that G/H is a torsion group.

23. T T T T T F T F T F

24. If $n \geq 2$, then $|A_n| = |S_n|/2$, so the only cosets of A_n are A_n and the set of all odd permutations in S_n. Thus the left and right cosets must be the same, and A_n is a normal subgroup of S_n. Because S_n/A_n has order 2, it is isomorphic to $\mathbb{Z}_2$. If $n = 1$, then $A_n = S_n$ so S_n/A_n is the trivial group of one element.

25. Let $h \in H$ and $a \in G$. Suppose left coset multiplication $(aH)(bH)$ by choosing representatives is well defined. Then $(a^{-1}H)(aH) = eH = H$. Choosing the representatives $a^{-1}h$ from $a^{-1}H$ and a from aH, we see that $a^{-1}ha = h_1$ for some $h_1 \in H$. Thus $ha = ah_1$, so $ha \in aH$. This shows that $Ha \subseteq aH$.

26. Exercise 39 of Section 11 proves that the elements of G of finite order do form a subgroup T of the abelian group G. Because G is abelian, every subgroup of G is a normal subgroup, so T is normal in G. Suppose that xT is of finite order in G/T; in particular, suppose that $(xT)^m = T$. Then $x^m \in T$. Because T is a torsion group, we must have $(x^m)^r = x^{mr} = e$ in G for some positive integer r. Thus x is of finite order in G, so that $x \in T$. This means that $xT = T$. Thus the only element of finite order in G/T is the identity T, so G/T is a torsion free group.

27. *Reflexive:* Because $i_e[H] = H$ for every subgroup H of G, we see that every subgroup is conjugate to itself.

Symmetric: Suppose that $i_g[H] = K$, so that for each $k \in K$, we have $k = ghg^{-1}$ for exactly one $h \in H$. Then $h = (g^{-1})kg = (g^{-1})k(g^{-1})^{-1}$, and we see that $i_{g^{-1}}[K] = H$, so K is also conjugate to H.

Transitive: Suppose that $i_a[H] = K$ and $i_b[K] = S$ for elements $a, b \in G$ and subgroups H, K, and S of G. Then each $s \in S$ can be written as $s = bkb^{-1}$ for a unique $k \in K$. But $k = aha^{-1}$ for a unique $h \in H$. Substituting, we have $s = b(aha^{-1})b^{-1} = (ba)h(a^{-1}b^{-1}) = (ba)h(ba)^{-1}$, so $i_{ba}[H] = S$ and H is conjugate to S.

28. We have $\overline{H} = \{H\}$ if and only if $gHg^{-1} = H$ for all $g \in G$, which is true if and only if H is a normal subgroup of G. We see that the normal subgroups of G are precisely the subgroups in the one-element cells of the conjugacy partition of the subgroups of G.

29. We see that $\rho_1\mu_2\rho_1^{-1} = \rho_1\mu_2\rho_2 = \mu_3$, and $\rho_2\mu_2\rho_2^{-1} = \rho_2\mu_2\rho_1 = \mu_1$, and conjugation by other elements of S_3 again yield either μ_1, μ_2, or μ_3. Thus the subgroups of S_3 conjugate to $\{\rho_0, \mu_2\}$ are $\{\rho_0, \mu_2\}, \{\rho_0, \mu_1\}$, and $\{\rho_0, \mu_3\}$.

30. We have $|G/H| = m$. Because the order of each element of a finite group divides the order of the group,we see that $(aH)^m = H$ for all elements aH of G/H. Computing using the representative a of aH, we see that $a^m \in H$ for all $a \in G$.

31. Let $\{H_i \mid i \in I\}$ be a set of normal subgroups of a group G. Let $K = \bigcap_{i\in I} H_i$. If $a, b \in K$, then $a, b \in H_i$ for each $i \in I$, and $ab \in H_i$ for each $i \in I$ because H_i is a subgroup of G. Thus $ab \in K$ and K is closed under the group operation of G. We see that $e \in K$ because $e \in H_i$ for each $i \in I$. Because $a^{-1} \in H_i$ for each $i \in I$, we see that $a^{-1} \in K$ also. Thus K is a subgroup of G. Let $g \in G$ and $k \in K$. Then $k \in H_i$ for $i \in I$ and $gkg^{-1} \in H_i$ for each $i \in I$ because each H_i is a normal subgroup of G. Hence $gkg^{-1} \in K$, and K is a normal subgroup of G.

32. Let $\{H_i \mid i \in I\}$ be the set of all normal subgroups of G containing S. Note that G is such a subgroup of G, so I is nonempty. Let $K = \bigcap_{i\in I} H_i$. By Exercise 31, we know that K is a normal subgroup of G, and of course K contains S because H_i contains S for each $i \in I$. By our constructions, we see that K is contained in *every* normal subgroup H_i of G containing S, so K must be the smallest normal subgroup of G containing S.

33. Consider two elements aC and bC in G/C. Now $(aC)^{-1} = a^{-1}C$ and $(bC)^{-1} = b^{-1}C$. Conseqently, choosing representatives, we see that $(aC)(bC)(aC)^{-1}(bC)^{-1} = aba^{-1}b^{-1}C$. However, $aba^{-1}b^{-1} \in C$ because C contains all commutators in G, so $(aC)(bC)(aC)^{-1}(bC)^{-1} = C$. Thus $(aC)(bC) = C(bC)(aC) = (bC)(aC)$ which shows that G/C is abelian.

34. Let $g \in G$. Because the inner automorphism $i_g : G \to G$ is a one-to-one map, we see that $i_g[H]$ has the same order as H. Because H is the only subgroup of G of that order, we find that $i_g[H] = H$ for all $g \in G$. Therefore H is invariant under all inner automorphisms of G, and hence is a normal subgroup of G.

35. By Exercise 54 of Section 5, we know that $H \cap N$ is a subgroup of G, and is contained in H, so it is a subgroup of H. Let $h \in H$ and $x \in H \cap N$. Then $x \in N$ and because N is a normal subgroup of G, we find that $hxh^{-1} \in N$, and of course $hxh^{-1} \in H$ because $h, x \in H$. Thus $hxh^{-1} \in H \cap N$, so $H \cap N$ is a normal subgroup of H.

Let $G = D_4$, let $N = \{\rho_0, \rho_2, \mu_1, \mu_2\}$, and let $H = \{\rho_0, \mu_1\}$, using the notation in Section 8. Then N is normal in G, but $H \cap N = H$ is not normal in G.

36. Let H be the intersection of all subgroups of G that are of order s. We are told that this intersection is nonempty. By Exercise 31, H is a subgroup of G. Let $x \in H$ and $g \in G$. Let K be any subgroup of G of order s. To show that $gxg^{-1} \in H$, we must show that $gxg^{-1} \in K$. Now $g^{-1}Kg$ is a subgroup of G of order s, so $x \in g^{-1}Kg$. Let $x = g^{-1}kg$ where $k \in K$. Then $k = gxg^{-1}$, so gxg^{-1} is indeed in K. Because K can be any subgroup of G of order s, we see that $gxg^{-1} \in H$, so H is a normal subgroup of G.

37. **a.** By Exercise 49 of Section 13, the composition of two automorphisms of G is a homomorphism of G into G. Because each automorphism is a one-to-one map of G onto G, their composition also has this property, and is thus an automorphism of G. Thus compostion gives a binary operation on the set of all automorphisms of G. The identity map acts as identity automorphism, and the inverse map of an automorphism of G is again an automorphism of G. Thus the automorphisms form a group under function composition.

b. For $a, b, x \in G$, we have $i_a(i_b(x)) = i_a(bxb^{-1}) = a(bxb^{-1})a^{-1} = (ab)x(b^{-1}a^{-1}) = (ab)x(ab)^{-1} = i_{ab}(x)$, so the composition of two inner automorphisms is again an inner automorphism. Clearly, i_e

acts as identity and the equation $i_a i_b = i_{ab}$ shows that $i_a i_{a^{-1}} = i_e$, so $i_{a^{-1}}$ is the inverse of i_a. Thus the set of inner automorphisms is a group under function composition.

Let $a \in G$ and let ϕ be any automorphism of G. We must show that $\phi i_a \phi^{-1}$ is an inner automorphism of G in order to show that the inner automorphisms are a normal subgroup of the entire automorphism group of G. For any $x \in G$, we have $(\phi i_a \phi^{-1})(x) = \phi(i_a(\phi^{-1}(x))) = \phi(a\phi^{-1}(x)a^{-1}) = \phi(a)\phi(\phi^{-1}(x))\phi(a^{-1}) = \phi(a)x(\phi(a))^{-1} = i_{\phi(a)}(x)$, so $\phi i_a \phi^{-1} = i_{\phi(a)}$ which is indeed an inner automorphism of G.

38. Let $H = \{g \in G \mid i_g = i_e\}$. Let $a, b \in H$. Then for $x \in G$, we have $(ab)x(ab)^{-1} = a(bxb^{-1})a^{-1} = axa^{-1} = x$, so $i_{ab} = i_e$ and $ab \in H$. Of course $e \in H$, and $axa^{-1} = x$ yields $x = a^{-1}xa = a^{-1}x(a^{-1})^{-1}$, so $a^{-1} \in H$. Thus H is a subgroup of G.

To show that H is a normal subroup of G, let $a \in H$ and $x \in G$. We must show that $xax^{-1} \in H$, that is, that $i_{xax^{-1}} = i_e$. For any $y \in G$, we have $i_{xax^{-1}}(y) = (xax^{-1})y(xax^{-1})^{-1} = x[a(x^{-1}yx)a^{-1}]x^{-1} = x(x^{-1}yx)x^{-1} = y = i_e(y)$, so $i_{xax^{-1}} = i_e$.

39. For $gH \in G/H$, let $\phi_*(gH) = \phi(g)H'$. Because we defined ϕ_* using the representative g of gH, we must show that ϕ_* is well defined. Let $h \in H$, so that gh is another representative of gH. Then $\phi(gh) = \phi(g)\phi(h)$. Because we are told that $\phi[H]$ is contained in H', we know that $\phi(h) = h' \in H'$, so $\phi(g)\phi(h) = \phi(g)h' \in \phi(g)H'$. This shows that ϕ_* is well defined, for the same coset $\phi(g)H'$ was obtained using the representatives g and gh.

For the homomorphism property, let $aH, bH \in G/H$. Because ϕ is a homomorphism, we obtain $\phi_*((aH)(bH)) = \phi_*((ab)H) = \phi(ab)H' = (\phi(a)\phi(b))H' = (\phi(a)H')(\phi(b)H') = \phi_*(aH)\phi_*(bH)$. Thus ϕ_* is a homomorphism.

40. a. Let H be the subset of $GL(n, \mathbb{R})$ consisting of the $n \times n$ matrices with determinant 1. The property $\det(AB) = \det(A)\cdot\det(B)$ shows that the set H is closed under matrix multiplication. Now $\det(I_n) = 1$ and every matrix in $GL(n, \mathbb{R})$ has a nonzero determinant and is invertible. From $1 = \det(I_n) = \det(AA^{-1}) = \det(A)\cdot\det(A^{-1})$, it follows that $\det(A^{-1}) = 1/\det(A)$, so if $A \in H$, then $A^{-1} \in H$. Thus H is a subgroup of $GL(n, \mathbb{R})$. Let $A \in H$ and let $X \in GL(n, \mathbb{R})$. Because X is invertible, $\det(X) \neq 0$. Then $\det(XAX^{-1}) = \det(X)\cdot \det(A)\cdot \det(X^{-1}) = \det(X)\cdot \det(A) \cdot (1/\det(X)) = \det(A) = 1$, so $XAX^{-1} \in H$. Thus H is a normal subgroup of $GL(n, \mathbb{R})$.

b. Let K be the subset of $GL(n, \mathbb{R})$ consisting of the $n \times n$ matrices with determinant ± 1. Note from Part(**a**) that if $\det(A) = -1$, then $\det(A^{-1}) = 1/(-1) = -1$. The same arguments as in Part(**a**) then show that if K is the subset of $n \times n$ matrices with determinant ± 1, then K is a subgroup of $GL(n, \mathbb{R})$. Part(**a**) shows that if $A \in K$ and $X \in GL(n, \mathbb{R})$, then $\det(XAX^{-1}) = \det(A) = \pm 1$, so that again $XAX^{-1} \in K$ and K is a normal subgroup of $GL(n, \mathbb{R})$.

41. a. Let A, B, and C be subsets of G. Then

$$\begin{aligned}(AB)C &= \{(ab)c \mid a \in A, b \in B, c \in C\} \\ &= \{a(bc) \mid a \in A, b \in B, c \in C\} = A(BC)\end{aligned}$$

by the associativity of multiplication in G. The subset $\{e\}$ acts as identity for this multiplication. Let $a, b \in G$ with $a \neq b$. Then the set $\{a, b\}$ has no multiplicative inverse, because the product of $\{a, b\}$ with any other nonempty subset of G yields a set with at least two elements, and hence not $\{e\}$. The product of any subset with the empty subset is the empty subset, so even if G = {e}, we still do not have a group, for $\varnothing$ has no inverse.